

Regression trees

Impurity as variance

- Bernoulli trial: probability of randomly labelling an example + is p . probability of labelling an example - is $1 - p$.
- Bernoulli distribution: $p(y = 1) = p, p(y = 0) = 1 - p$.
Expectation: $E(y) = p$
Variance: $\sigma^2 = p(1 - p)$
- Gini index $2p(1 - p)$ viewed as a variance term.
- Partition of $D = \{D_1, D_2\}$, $|D| = n$, $|D_1| = n_1$, $|D_2| = n_2$
Partition impurity =
$$\frac{n_1}{n}2p(1 - p) + \frac{n_2}{n}2p(1 - p) = 2 \left(\frac{n_1}{n}\sigma_1^2 + \frac{n_2}{n}\sigma_2^2 \right)$$
- Partition impurity is small $\Rightarrow$ each segment has small variance.

Minimize variances of segments

- Partition $(Y) = \{Y_1, \dots, Y_L\}$.

Partition variance, $\text{Var}\{Y_1 \dots Y_L\}$

$$= \sum_{i=1}^L \frac{|Y_i|}{|Y|} \text{Var}(Y_i)$$

$$= \sum_{i=1}^L \frac{|Y_i|}{|Y|} \left(\frac{1}{|Y_i|} \sum_{y \in Y_i} y^2 - \bar{y}_i^2 \right)$$

$$= \frac{1}{|Y|} \sum_{y \in Y} y^2 - \sum_{i=1}^L \frac{|Y_i|}{|Y|} \bar{y}_i^2$$

- Minimize partition variance $\equiv$ Maximize the weighted average of squared means, $\sum_{i=1}^L \frac{|Y_i|}{|Y|} \bar{y}_i^2$ of the segments.

Best feature for partitioning the data set

Algorithm: BestFeature (D, F)

input : Data set D , features F

output: Best feature f to partition D

```
1 Variancemin  $\leftarrow$  1
2 foreach  $f \in F$  do
3      $\{D_1 \dots D_L\} \leftarrow$  Partition  $D$  based on  $f$  values.
4     if Variance  $\{D_1 \dots D_L\} <$  Variancemin then
5         Variancemin  $\leftarrow$  Variance $\{D_1 \dots D_L\}$ 
6          $f_{best} \leftarrow f$ 
7     end
8 end
9 return  $f_{best}$ 
```

Label and homogeneous

- $\text{Label}(Y) = \text{mean value of } Y$
- $\text{Homogeneous}(Y) = \text{true if variance}(Y) = 0, \text{ or less than a threshold.}$

Regression tree construction

Algorithm: RegressionTree (D, F)

input : Data set D , features F

output: Regression tree T with labelled leaves

```
1 if Homogeneous( $D$ ) then return Label( $D$ )
2  $f \leftarrow$  BestFeature ( $D, F$ )
3 Partition  $D$  into  $\{D_1 \dots D_L\}$  based on the literals  $f = v$ 
4 foreach  $D_i$  do
5     if  $D_i \neq \emptyset$  then  $T_i \leftarrow$  DecisionTree ( $D_i, F$ )
6     else  $T_i$  is a leaf labelled with Label ( $D$ )
7 end
8 return a tree whose root is labelled with  $f$  and whose
   children are  $T_i$ 
```

Learning a regression tree

#	Model	Condition	Leslie	Price
1.	B3	excellent	no	4513
2.	T202	fair	yes	625
3.	A100	good	no	1051
4.	T202	good	no	270
5.	M102	good	yes	870
6.	A100	excellent	no	1770
7.	T202	fair	no	99
8.	A100	good	yes	1900
9.	E112	fair	no	77

Model: B3 = 1, T202 = 2, A100 = 3, M102 = 4, E112 = 5

Condition: fair = 1, good = 2, excellent = 3

Leslie: no = 0, yes = 1

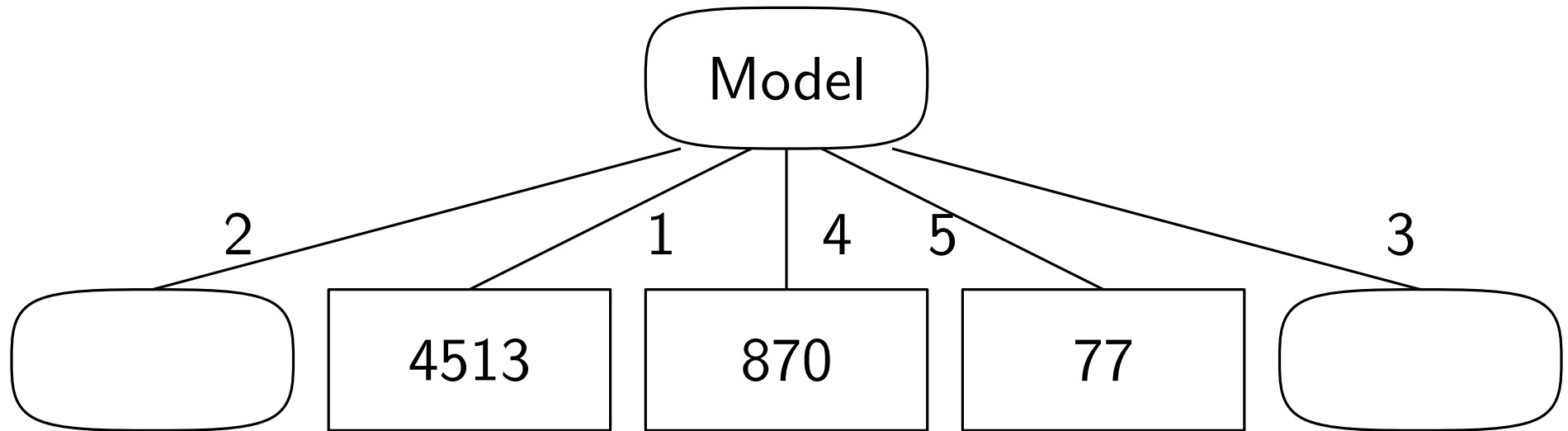
Learning a regression tree

#	M	C	L	P		M	1	[4513]
1.	1	3	0	4513			2	[625, 270, 99]
2.	2	1	1	625			3	[1051, 1770, 1900]
3.	3	2	0	1051			4	[870]
4.	2	2	0	270			5	[77]
5.	4	2	1	870	C		1	[625, 99, 77]
6.	3	3	0	1770			2	[1051, 270, 870, 1900]
7.	2	1	0	99			3	[4513, 1770]
8.	3	2	1	1900	L		0	[4513, 1051, 270, 1770, 99, 77]
9.	5	1	0	77			1	[625, 870, 1900]

Learning a regression tree

		Partition	Mean	Squared means
M	1	[4513]	4513	
	2	[625, 270, 99]	331	
	3	[1051, 1770, 1900]	1574	
	4	[870]	870	
	5	[77]	77	$3.21 \cdot 10^6$
C	1	[625, 99, 77]	267	
	2	[1051, 270, 870, 1900]	1023	
	3	[4513, 1770]	3142	$2.68 \cdot 10^6$
L	0	[4513, 1051, 270, 1770, 99, 77]	1297	
	1	[625, 870, 1900]	1132	$1.55 \cdot 10^6$

Regression tree



Learning a regression tree

#	M	C	L	P
1.	1	3	0	4513
2.	2	1	1	625
3.	3	2	0	1051
4.	2	2	0	270
5.	4	2	1	870
6.	3	3	0	1770
7.	2	1	0	99
8.	3	2	1	1900
9.	5	1	0	77

#	M	C	L	P
2.	2	1	1	625
3.	3	2	0	1051
4.	2	2	0	270
6.	3	3	0	1770
7.	2	1	0	99
8.	3	2	1	1900

Learning a regression tree

#	M	C	L	P
2.	2	1	1	625
3.	3	2	0	1051
4.	2	2	0	270
6.	3	3	0	1770
7.	2	1	0	99
8.	3	2	1	1900

M = 2		
C	1	[625,99]
	2	[270]
	3	[]
L	0	[270,99] 185
	1	[625] 625

M=3		
C	1	[]
	2	[1051,1990]
	3	[1770]
L	0	[1051,1770] 1411
	1	[1900] 1900

Regression tree

